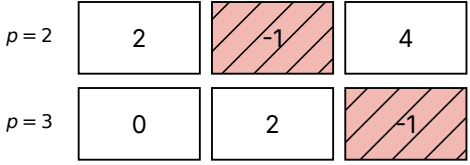
**a** valuation matrix (hatched: negative entries)

$V_A = (\Delta_p(N_j))$, rows $p = 2, 3$; columns $N = 1, 2, 3$




$V_A\,(7, -2, -4)^{\mathsf{T}} = (0, 0)^{\mathsf{T}}$, $\ker_{\mathbb{Z}} V_A = \mathbb{Z}(7, -2, -4)$

**b** the multiplier identity

$$R(1)^7 = R(2)^2 R(3)^4 = 4^7$$

$R(1) = 4,\ R(2) = \frac{9}{2},\ R(3) = \frac{16}{3}$

$(\frac{9}{2})^2 (\frac{16}{3})^4 = \frac{81}{4} \cdot \frac{65536}{81} = 16384$

source 4 → endpoint 65536